Scaffolding Authentic Inquiry in Engineering Laboratories via Smartphone Cross-Modal Signal Experiments: A Process Traceability Framework

Empirical Evaluation of Cognitive Reorientation, Valid Inquiry Loops, and ABET Student Outcomes Across Three Longitudinal Cohorts

Abstract—Traditional communication and signal processing laboratories often suffer from prohibitive instrument costs, rigid spatial-temporal schedules, and passive 'cookbook-style' procedural verifications. Although Bring Your Own Device (BYOD) smartphone experimentation presents a pervasive alternative, mobile devices frequently induce attentional fragmentation due to pervasive digital distractions. This article presents an inquiry-based engineering pedagogical framework centered on counterintuitive physical phenomena across optical, acoustic, and radio-frequency (RF) modalities, supported by a lightweight local area network (LAN) platform. Utilizing smartphone embedded CMOS rolling shutter cameras for visible light communication (VLC), microphones for acoustic pulse-compression echo ranging, and the phone as a zero-install LAN client paired with dedicated dual-ESP32 nodes for Wi-Fi Channel State Information (CSI) sensing, students investigate spatial optical fringes, acoustic multipath interference, and RF Doppler dynamics within a structured Predict-Observe-Explain-Challenge-Redesign (POE+CR) cycle. The supporting platform implements a structured 'Run Record' directed acyclic graph (DAG) model, automated pre-flight environment baselining, single-variable Diff View visualization, and a Valid Inquiry Loop decision engine that reduces unconstrained guess-and-check manipulation. A longitudinal three-year quasi-experimental study across 288 undergraduate engineering students (N = 288, 72 four-person teams) demonstrates that the experimental inquiry cohort achieved 1.80 times as many valid inquiry loops per unit as the control cohort (M = 4.35 vs. 2.42, $t(214) = 18.16$, $p < 0.001$, Cohen's $d = 2.47$), an 11.20-point increase in post-test conceptual mastery (M = 88.6 vs. 77.4, $t(286) = 13.04$, $p < 0.001$, Cohen's $d = 1.55$), and substantially higher attainment rates across ABET Student Outcomes 1, 5, and 6 ($p < 0.001$). These findings suggest that structuring mobile experiments around counterintuitive physical discrepancies and transparent process lineage can help reorient smartphones from digital distractions toward constructive cognitive scaffolds for authentic engineering inquiry.

Keywords—Engineering education, Smartphone-based experiments, Cross-modal sensing, Inquiry-based learning, Process traceability, ABET accreditation.

1. INTRODUCTION

Laboratory education forms the empirical cornerstone of undergraduate engineering curricula, bridging theoretical mathematical formulations with non-ideal physical realities [8]. In electrical and computer engineering (ECE) programs, courses in communication theory, digital signal processing (DSP), and the Internet of Things (IoT) depend fundamentally on experiential laboratory work to anchor concepts such as Fourier spectral decomposition, spatial electromagnetic wave propagation, and baseband modulation schemes [9, 10]. However, engineering departments worldwide face substantial structural bottlenecks in laboratory delivery. Commercial desktop instruments—such as high-bandwidth oscilloscopes, vector network analyzers, and RF signal generators—entail capital expenditures ranging from $50,000 to $100,000 per laboratory station, imposing severe budgetary constraints on institutional expansion [10, 11]. Furthermore, conventional physical laboratories are inherently confined by fixed spatial-temporal timetables, limiting students to bi-weekly, tightly scheduled sessions that leave little latitude for open-ended exploration [11, 12].

To mitigate these physical barriers, mobile learning paradigms leveraging students' own devices—commonly termed Bring Your Own Device (BYOD)—have garnered significant attention across international engineering education [1, 2]. Modern commercial smartphones integrate high-grade complementary metal-oxide-semiconductor (CMOS) image sensors, high-fidelity multi-microphone audio subsystems, and advanced multi-radio wireless chipsets. Pioneering platforms such as phyphox have demonstrated that smartphone sensors can transform personal mobile devices into portable science and engineering instruments [3, 4]. These mobile experimental toolsets have been shown to elevate situational interest and foster autonomous empirical exploration in physics and foundational STEM courses [4, 5].

Nonetheless, deploying smartphones as primary experimental apparatuses in engineering classrooms presents an acute pedagogical dilemma. Unlike specialized benchtop hardware, smartphones function as hyper-converged communication hubs that continuously stream non-academic notifications from social media, video platforms, and instant messaging services [6]. Rigorous cognitive load and psychological research demonstrates that the mere physical presence of a personal smartphone extracts substantial working memory resources—a cognitive penalty termed 'brain drain' [7]. In unstructured BYOD environments, this attentional conflict frequently results in task disengagement, surface-level interaction, and fragmented inquiry, undermining the cognitive focus required to master rigorous engineering signal concepts.

Compounding this attentional challenge is the 'black-box' dilemma inherent in many contemporary educational laboratory software suites. Both remote online laboratories and virtual simulation packages predominantly focus on cloud-based instrument virtualization and graphical interface distribution [10, 11, 12]. When students encounter non-ideal experimental waveforms or failed performance thresholds, typical software interfaces fail to scaffold systematic troubleshooting. Instead, learners frequently resort to unconstrained multi-variable 'guess-and-check' manipulation or rapid browser refreshes. Because standard platforms record only final static outputs rather than the intermediate parametric lineage, educators cannot evaluate whether students followed controlled-variable methodologies or attained principled physical reasoning. This opacity impedes authentic assessment of ABET student outcomes, particularly Outcome 1 (Complex Problem Solving) and Outcome 6 (Experimental Design and Data Interpretation) [23, 24].

To reconcile this tension between ubiquitous mobile sensing and digital distraction, this study investigates an alternative pedagogical and technological paradigm: Can smartphones be purposefully restructured into rigorous cognitive scaffolds for authentic engineering inquiry? Specifically, we formulate three core research questions (RQs):

- **RQ1 (Pedagogical Mechanism):** How can counterintuitive physical signal phenomena and structured hypothesis-testing cycles be operationalized to reorient student attention from digital distraction toward sustained epistemic curiosity?
- **RQ2 (Process Traceability Architecture):** How can a lightweight web-based laboratory platform scaffold single-variable controlled experimentation and transparent process lineage across heterogeneous student smartphones without requiring native client installations?
- **RQ3 (Empirical Efficacy):** What quantitative impacts does this inquiry-based framework produce on students' valid inquiry loop frequencies, conceptual knowledge acquisition, and ABET Student Outcomes attainment over longitudinal academic cohorts?

To address these questions, we designed and deployed a comprehensive cross-modal signal experiment suite—spanning visible light optical camera communication (VLC/OCC), acoustic pulse-compression echo ranging, and RF Wi-Fi Channel State Information (CSI) sensing—supported by a lightweight local area network (LAN) platform. Over three academic years (2023–2026), we conducted a controlled quasi-experimental study across 288 undergraduate engineering students. The remainder of this paper is organized as follows: Section 2 establishes the pedagogical model and ABET rubric integration; Section 3 details the physical experiment suite; Section 4 articulates the platform software architecture; Section 5 presents inferential statistical evaluations; Section 6 discusses pedagogical and operational implications; and Section 7 concludes the paper.

2. PEDAGOGICAL FRAMEWORK AND LEARNING DESIGN

2.1 Theoretical Foundations: Epistemic Curiosity and Counterintuitive Discrepancies

Our instructional design is grounded in cognitive constructivism and Daniel Berlyne's theory of epistemic curiosity—the intrinsic motivational drive to resolve perceptual incongruity and conceptual uncertainty [13]. Cognitive neuroscience demonstrates that activating epistemic curiosity stimulates midbrain dopaminergic reward pathways and enhances hippocampal long-term memory consolidation [14]. When students encounter counterintuitive physical phenomena that directly contradict intuitive expectations, cognitive equilibrium is disrupted. This 'expectation violation' creates a compelling learning opportunity, encouraging learners to formulate and evaluate physical hypotheses [14, 15].

To systematically harness this cognitive tension, we adapt inductive teaching paradigms [16] and cognitive load theory [17]. When intrinsic cognitive load is carefully managed through structured instructional scaffolding, working memory can be fully dedicated to germane schema construction [17, 19]. Self-determination theory further posits that autonomous, competence-building empirical exploration fosters deep academic engagement [18]. By guiding students to confront real-world engineering ambiguities [20], challenge-based inquiry transforms passive procedural verification into authentic engineering problem solving [21, 22].

2.2 The Unified Cross-Modal Signal Processing Chain

A foundational challenge in undergraduate ECE education is curricular fragmentation: students often encounter optical, acoustic, and electromagnetic signals in separate courses, failing to appreciate their shared mathematical foundation. To cultivate holistic engineering thinking, we synthesize these disparate physical domains into a universal five-stage signal processing chain: Source Modulation -> Channel Dynamics -> Transducer Sampling -> Feature Extraction -> Decision/Demodulation. In every experimental module, regardless of physical carrier, students interrogate the exact same structural questions: How does emitter modulation format information? How do channel multipath and path loss distort waveforms? How do sensor sampling constraints govern signal recovery? How are discriminative features extracted from raw time-series data? And how are robust statistical decision boundaries established?

2.3 The Sequential POE+CR Five-Stage Evidence Cycle

Expanding classical Predict-Observe-Explain (POE) models [15] with Challenge and Redesign extensions [21], we structure each 180-minute laboratory session into a strict five-stage evidence

cycle: Predict (E1) -> Observe (E2) -> Explain (E3) -> Challenge (E4) -> Redesign (E5). Each stage enforces a structured evidence artifact that is recorded within the platform's lineage database:

- **Stage E1 (Predict - Prior Hypothesis):** Before activating physical emitters, students must mathematically model the expected signal behavior and record quantitative predictions regarding waveform morphologies, frequency spectra, or spatial bounds.
- **Stage E2 (Observe - Baseline Acquisition):** Students initialize physical hardware, execute automated ambient noise baselining, capture multi-frame raw sensor signals, and document uncorrupted baseline features.
- **Stage E3 (Explain - Mechanistic Reconciliation):** Students extract analytical features from raw waveforms, reconcile empirical observations with their Stage E1 predictions, and mathematically articulate any observed discrepancies.
- **Stage E4 (Challenge - Controlled Parameter Exploration):** Students receive a counterintuitive physical stress scenario (e.g., severe multipath reflections, optical underexposure, or channel spatial shifts). They must execute single-variable parameter iterations, systematically adjusting one parameter per run while tracking parameter effects.
- **Stage E5 (Redesign - Robust Engineering System Synthesis):** Students synthesize experimental insights to formulate a structural system redesign—proposing modified modulation parameters, adaptive filtering pipelines, or receiver algorithms to permanently overcome observed channel impairments.

2.4 Alignment with ABET Student Outcomes and Rubric Mapping

To ensure rigorous curricular quality control, our pedagogical framework maps each POE+CR stage directly to international ABET Engineering Accreditation Commission criteria [23], specifically Outcomes 1, 5, and 6 [24, 25]:

- **ABET SO 1 (Complex Problem Solving):** Addressed in Stages E3 and E5, requiring students to identify, formulate, and solve complex engineering problems by applying mathematical and engineering principles to resolve physical signal distortions.
- **ABET SO 5 (Collaborative Teamwork):** Addressed through structured four-role collaborative team rotations, fostering collaborative leadership, task accountability, and peer consensus during experimental iterations [26, 27].
- **ABET SO 6 (Rigorous Experimentation and Data Analysis):** Addressed in Stages E1, E2, and E4, requiring students to develop appropriate experimentation, analyze raw multi-channel data, and employ engineering judgment to draw valid conclusions [28].

Table 1. Core Pedagogical Design Principles of the Cross-Modal Inquiry Framework

Design Principle	Theoretical Grounding	Operational Implementation	Target Educational Objective
1. Phenomenological Discrepancy	Epistemic Curiosity [13, 14]	Present counterintuitive physical phenomena at lab outset (e.g., optical stripes, acoustic ghost peaks)	Transform habitual smartphone interaction into intrinsic investigative motivation

Design Principle	Theoretical Grounding	Operational Implementation	Target Educational Objective
2. Unified Signal Abstraction	Cognitive Constructivism [16, 19]	Map optical, acoustic, and RF modalities onto identical Source-Channel-Sampling-Feature-Decision chain	Cultivate cross-disciplinary systems thinking and break down course knowledge silos
3. Strict Variable Isolation	Scientific Inquiry & Metacognition [20, 28]	Software-enforced single-parameter modification rule (Diff View verification)	Reduce chaotic trial-and-error; instill disciplined engineering experimental habits
4. End-to-End Process Traceability	Process Transparency & Evidence Theory [24]	Structured parent-child Run Record DAG model recording hardware states and waveforms	Mitigate data fabrication; enable verifiable audit trails for ABET accreditation

Table 2. Alignment Between POE+CR Inquiry Stages and ABET Student Outcomes (SOs)

ABET Student Outcome [23]	Core Competency Definition	Associated POE+CR Stages	Evidence Artifact & Evaluation Metric
SO 1: Complex Problem Solving	Ability to identify, formulate, and solve complex engineering problems using engineering, science, and math principles	Stage E3 (Mechanistic Explanation) Stage E5 (System Redesign Proposal)	Mathematical rigor in reconciling expectation violations (20 pts); engineering feasibility and robustness of Stage E5 redesign (15 pts)
SO 5: Collaborative Teamwork	Ability to function effectively on a team whose members together provide leadership and create a collaborative environment	Continuous Across All Stages (Role Rotation in E1–E5)	Adherence to four-role operational protocols (Operator, Recorder, Analyst, Auditor); intra-team peer evaluation rubrics (15 pts)
SO 6: Experimental Design & Data Interpretation	Ability to develop and conduct appropriate experimentation, analyze and interpret data, and draw valid engineering conclusions	Stage E1 (Hypothesis Modeling) Stage E2 (Baseline Acquisition) Stage E4 (Controlled Challenge Loops)	Hypothesis predictive validity (15 pts); baseline waveform integrity (15 pts); frequency and validity of controlled inquiry loops (20 pts)

2.5 Collaborative Team Dynamic: Four-Role Rotational Accountability

To ensure equitable participation and cultivate professional teamwork, students operate in four-member collaborative teams governed by structured rotational roles [25, 26]: (1) Primary Hardware Operator (manages physical emitters, sensors, and spatial alignments); (2) Data & Telemetry Recorder (supervises baseline calibration, triggers data acquisitions, and logs platform timestamps); (3) Signal Analyst (computes mathematical features, generates spectral representations, and leads Stage E3 explanations); and (4) Quality Auditor & Systems Designer (enforces single-variable control, verifies parameter diffs, and synthesizes Stage E5 redesigns). Roles rotate across the three physical

modules, guaranteeing that every student masters hands-on hardware operation, software-based data interpretation, and high-level engineering synthesis.

3. CROSS-MODAL SIGNAL EXPERIMENT SUITE DESIGN

The experimental curriculum encompasses three complementary physical signal sensing modalities: Visible Light Optical Camera Communication (VLC/OCC), Acoustic Pulse-Compression Echo Ranging (Sonar), and Radio-Frequency Wi-Fi Channel State Information (CSI) Motion Sensing. Table 3 summarizes the comparative technical specifications across the three modules.

Table 3. Comparative Technical and Architectural Matrix of the Three Cross-Modal Signal Modules

Physical Parameter	Visible Light (VLC/OCC)	Acoustic Ranging (Sonar)	Radio-Frequency (Wi-Fi CSI)
Physical Medium & Wave Type	Electromagnetic Light Wave (400–790 THz)	Longitudinal Mechanical Pressure Wave	Electromagnetic RF Wave (2.4 GHz ISM)
Propagation Speed in Air	$c \approx 3.0 \times 10^8$ m/s	$c_s \approx 343$ m/s (at 20°C)	$c \approx 3.0 \times 10^8$ m/s
Signal Bandwidth & Modulation	Square-wave PWM 2FSK (250 Hz / 500 Hz)	Linear Chirp (17–20 kHz, B = 3 kHz)	802.11n OFDM (52 active subcarriers, 20 MHz)
Primary Sensor & Interface	CMOS Rolling Shutter Camera (30 fps)	Audio Microphone (44.1 kHz, 16-bit)	Dual ESP32 Nodes (Phone as LAN Web Client)
Counterintuitive Phenomenon	Continuous LED appears as spatial zebra stripes	Acoustic cross-correlation displays false ghost peaks	Body movement at 3 m shifts RF subcarrier amplitudes
Governing Analytical Formulation	$P = 1 / (f_{PWM} \cdot t_r)$ [Eq. 1]	$d_{est} = c_s \cdot \Delta\tau / 2$ [Eq. 2]	$a_k(t) = \sqrt{I_k(t)^2 + Q_k(t)^2}$ [Eq. 3]
Controlled Inquiry Parameter (E4)	Camera exposure time t_{exp}, distance, angle	Chirp duration, windowing, multipath reflectors	Subcarrier index selection, motion speed, link geometry

3.1 Module 1: Visible Light Optical Camera Communication (VLC/OCC)

In Module 1, an LED modulated by an open-source ESP32 microcontroller emits a 2FSK optical signal (250 Hz for bit '0', 500 Hz for bit '1', 50% duty cycle, symbol duration 250 ms) [33, 34]. When viewed with the human eye, the high-frequency modulation blends into continuous, flicker-free white illumination. However, when students capture the LED using their smartphone's CMOS camera, the rolling shutter mechanism exposes sensor scanlines sequentially at line readout interval t_r (typically 15–35 µs depending on smartphone model) [35]. The temporal intensity variation is thus mapped onto a spatial striping pattern across video frames. The spatial stripe period P (measured in vertical pixels) satisfies:

$$P = 1 / (f_{PWM} \cdot t_r) \quad (1)$$

Students observe distinct spatial frequencies corresponding to binary states: wider dark-bright zebra stripes for 250 Hz and dense stripes for 500 Hz. In Stage E4, students encounter intentional underexposure and ambient optical interference. They must calibrate row readout time t_r across

varying exposure durations, demonstrating how rolling shutter spatial sampling circumvents native frame-rate limits (30 fps) to achieve high-frequency optical signal detection.

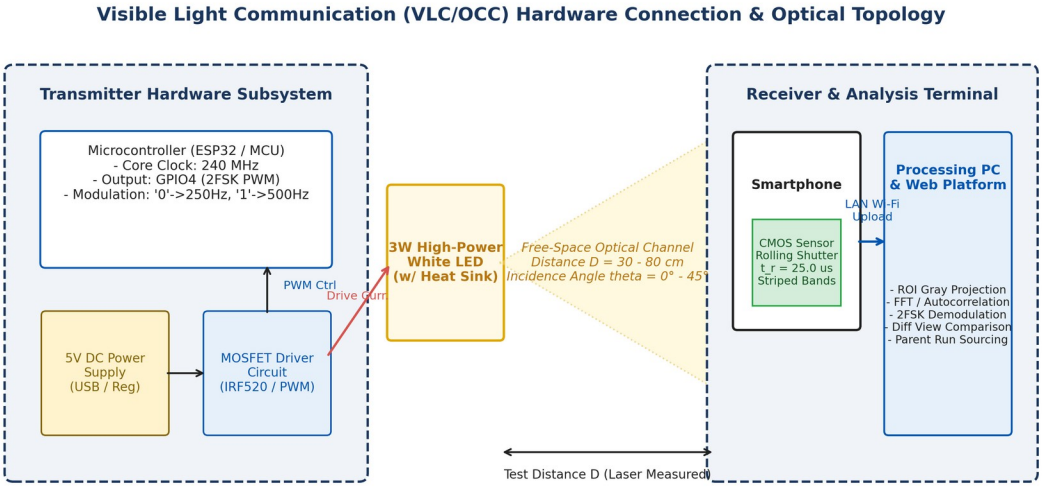


Figure 1. Hardware topology, optical transmission link, and CMOS rolling shutter spatial striping in Module 1 (VLC/OCC).

3.2 Module 2: Acoustic Pulse-Compression Echo Ranging (Sonar)

Module 2 explores mechanical wave propagation and matched filtering. A dedicated audio transducer emits a near-ultrasound linear frequency modulated (LFM) chirp pulse s(t) sweeping from f_0 = 17 kHz to f_1 = 20 kHz over duration T = 20 ms (effective acoustic bandwidth B = 3 kHz) [36]. The smartphone microphone captures the direct acoustic transmission and reflected echo signals r(t). In the web platform, the digitized audio stream undergoes cross-correlation with the matched reference filter: $R_{sr}(\tau) = \int s(t)\, r(t+\tau)\, dt$. The target range d_est is estimated from the round-trip time-of-flight (TOF) peak Δτ [37, 38]:

$$d_est = (c_s \cdot \Delta\tau) / 2 \quad (2)$$

where c_s ≈ 343 m/s represents the speed of sound in air at 20℃. In the laboratory environment, acoustic reflections from desk surfaces and adjacent walls introduce prominent multipath interference, generating spurious correlation peaks ('ghost echoes'). Students learn that with acoustic bandwidth B = 3 kHz, the theoretical spatial resolution limit is $\Delta R = c_s / (2B) \approx 343 / (2 \times 3000) \approx 5.7$ cm. Consequently, physical obstacles separated by less than 5.7 cm cannot be resolved as distinct peaks without advanced deconvolution. In Stage E4, students investigate multipath mitigation by varying chirp durations, applying Kaiser windowing functions, and repositioning acoustic baffles.

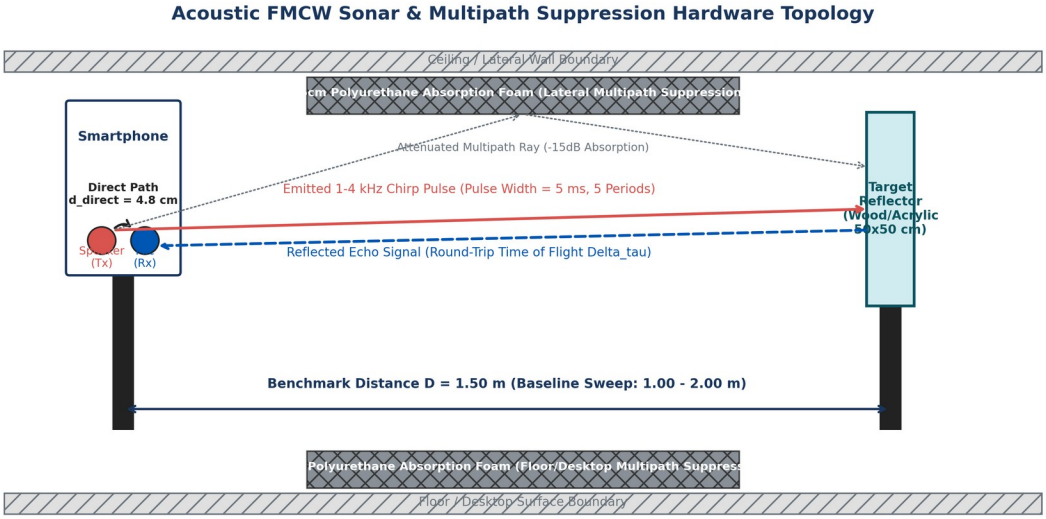


Figure 2. Acoustic echo ranging topology, LFM chirp transmission, and matched-filter cross-correlation TOF peak detection in Module 2.

3.3 Module 3: Radio-Frequency Wi-Fi Channel State Information (CSI) Sensing

Module 3 extends signal principles to radio-frequency propagation and spatial Fresnel zone dynamics [39, 40]. Modern IEEE 802.11n/ac Wi-Fi systems utilize Orthogonal Frequency Division Multiplexing (OFDM). The physical wireless channel across subcarrier k is mathematically represented by its complex Channel State Information: H_k = I_k + j Q_k. The instantaneous subcarrier amplitude a_k(t) is extracted as [41, 42]:

$$a_k(t) = sqrt(I_k(t)^2 + Q_k(t)^2) \quad (3)$$

To circumvent mobile operating system sandboxing—which strictly isolates raw baseband CSI from mobile web browsers—we deploy dedicated low-cost ESP32 nodes configured as an active RF transceiver link spaced at 3.0 m Line-of-Sight (LoS). The receiver node logs raw CSI complex matrices across 52 subcarriers at 100 Hz, streaming processed spatial variance packets to the laboratory web server. Students use their personal smartphones as zero-install LAN web clients to monitor real-time subcarrier dynamics. When a human subject traverses the line of sight, dynamic path length variations induce multi-path phase shifts, causing substantial amplitude fluctuations across specific subcarriers. In Stage E4, students calibrate motion detection thresholds and select optimal subcarrier masks to discriminate human locomotion from static ambient noise.

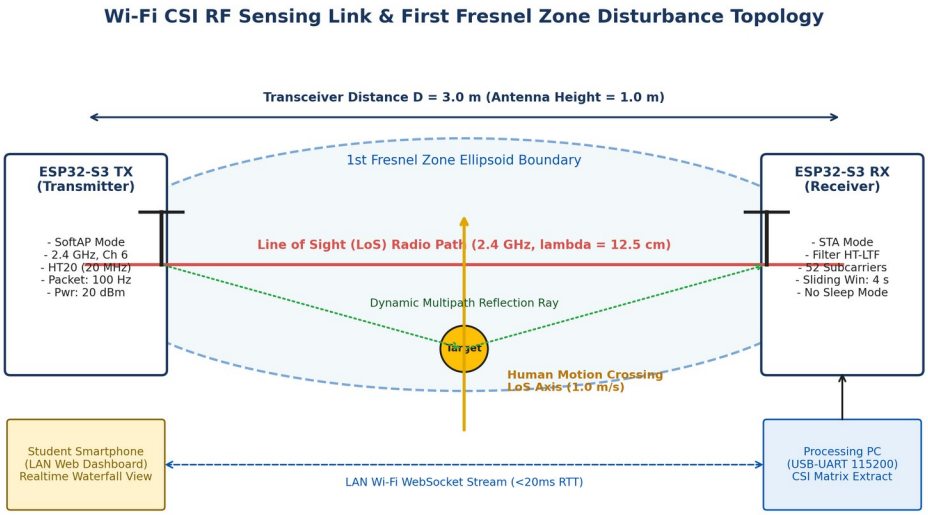


Figure 3. RF Wi-Fi CSI sensing topology, dedicated dual-ESP32 transceiver link (3.0 m LoS), and subcarrier amplitude dynamics in Module 3.

Table 4. Standard 180-Minute Laboratory Instructional Protocol and Time Allocation

Phase / Stage	Duration	Instructional & Student Activities	Software Scaffolding & Enforcement	Mandatory Evidence Artifact
Orientation & Setup	0–20 min	Instructor introduces counterintuitive phenomenon; teams inspect hardware and assign roles	Platform displays module challenge brief and enforces team roster binding	Verified team login & assigned role sheet
E1: Prior Prediction	20–40 min	Students model expected signal response and record quantitative mathematical predictions	Platform locks data acquisition buttons until Stage E1 hypothesis form is submitted	Stage E1 Hypothesis & Mathematical Form
E2: Baseline Acquisition	40–75 min	Hardware power-on; automated environment pre-flight baselining; capture multi-frame raw signals	Pre-flight baselining engine evaluates ambient noise floor; triggers raw wave archival	Baseline Waveform & Noise Floor Profile
E3: Mechanistic Explanation	75–105 min	Feature extraction; waveform spectral decomposition; reconcile data with Stage E1 predictions	Diff View plots overlay theoretical expectations against empirical sensor traces	Stage E3 Reconciled Mechanistic Analysis

Phase / Stage	Duration	Instructional & Student Activities	Software Scaffolding & Enforcement	Mandatory Evidence Artifact
E4: Controlled Challenge	105–150 min	Students execute single-variable iterations under environmental stress constraints	Valid Inquiry Loop engine blocks runs if multiple parameters are altered simultaneously	Structured Inquiry DAG & Diff Trace Ledger
E5: System Redesign	150–180 min	Synthesize findings; propose robust engineering redesign to overcome channel vulnerabilities	Platform opens engineering proposal form supporting code snippets and schematic uploads	Stage E5 Engineering System Redesign Proposal

4. LIGHTWEIGHT LABORATORY PLATFORM ARCHITECTURE

4.1 Local LAN Deployment and Zero-Install Web Client

To avoid installation friction, device operating system incompatibilities, and cloud dependency, the laboratory platform is engineered as a zero-install, lightweight web application deployed entirely on an offline local area network (LAN). The server runs on a single low-cost host computer (or Raspberry Pi 4) executing Python FastAPI with SQLite stored in a high-speed tmpfs in-memory cache. Students connect their personal smartphones to the laboratory Wi-Fi router and access the responsive HTML5/Canvas interface via any standard mobile browser (Safari, Chrome, Firefox, Edge). Real-time telemetry, live spectrograms, and raw waveform streams are pushed bidirectionally via lightweight WebSockets, achieving sub-10 ms rendering latency on student devices [29, 30].

4.2 'Run Record' Lineage and DAG Process Modeling

The core data architecture is structured around a directed acyclic graph (DAG) model of experimental runs. Every parameter submission, data acquisition, and analytical notation is bundled into a discrete 'Run Record' containing: (1) a parent Run ID establishing parameter lineage; (2) a complete snapshot of all hardware transmitter and software receiver parameters; (3) the raw multi-modal time-series waveform files; (4) extracted mathematical feature vectors; and (5) a SHA-256 cryptographic digest binding parameters, waveforms, and student explanations. This DAG structure provides instructors with a transparent, step-by-step audit trail of each student team's exploratory process.

4.3 Pre-flight Baseline Audit and Valid Inquiry Loop Decision Engine

To dismantle the 'guess-and-check' trial-and-error approach common in unstructured lab environments, the platform embeds a real-time Valid Inquiry Loop decision engine. When a student team initiates a new experimental run from an established baseline parent, the engine compares parameter vectors across the two runs. If the Hamming distance between parameter vectors satisfies $\Delta k = 1$, and the team records an explicit physical justification prior to execution, the run is validated and cataloged as a 'Valid Inquiry Loop'. If a team attempts to alter multiple parameters

simultaneously (Δk ≥ 2), the system flags the submission with a controlled-variable warning, requiring students to either isolate a single variable or supply a rigorous multi-factor justification before data capture proceeds (Table 5).

Table 5. Architectural Decision Rules for the Valid Inquiry Loop Engine

Operational Condition	Algorithmic Decision Criteria	System Response & Enforcement	Pedagogical Rationale
Controlled Single-Variable Run	Parameter diff: Δk = 1; valid text justification provided	Approve execution; record as Valid Inquiry Loop; increment team progress counter	Reinforces rigorous scientific isolation of variables and deliberate inquiry habits
Uncontrolled Multi-Variable Run	Parameter diff: Δk ≥ 2 between successive runs	Halt execution; highlight confounded parameters in Diff View; require isolation	Discourages unconstrained 'guess-and-check' trial-and-error; fosters physical reflection
No Parameter Variation (Clone Run)	Parameter diff: Δk = 0; repeated identical run	Issue prompt: 'Repeat run detected. Are you evaluating stochastic variance?'	Discourages blind interface clicking while preserving intentional repeatability tests
Missing Baseline Audit	Parent Run ID = NULL or baseline environment unverified	Block challenge runs; redirect team to Stage E2 ambient baselining	Ensures all experimental deductions are anchored against calibrated physical ground truth

4.4 Parameter Diff View and Live Teacher Telemetry Dashboard

Inspired by software version control, the platform provides an interactive 'Diff View' that visually renders parameter differences between any two runs in parallel columns, synchronizing corresponding time-domain waveforms and frequency spectra. Simultaneously, the teacher telemetry dashboard provides instructors with real-time classwide visibility. Instructors can instantly identify teams stuck in invalid loops, view distribution heatmaps of inquiry activity, and intervene proactively with targeted Socratic questions.

4.5 Device Heterogeneity Mitigation and Signal Normalization

A notorious obstacle in BYOD laboratory implementations is device heterogeneity—differences in camera sensor dimensions, rolling shutter line readout speeds, microphone acoustic frequency responses, and operating system audio filters [31, 32]. Our platform resolves this challenge through four cross-platform normalization mechanisms: (1) CMOS line readout calibration routines that dynamically compute scanline intervals t_r via two-frequency reference LED bursts; (2) Web Audio API configurations that programmatically disable OS-level acoustic echo cancellation (AEC), automatic gain control (AGC), and noise suppression; (3) rational polynomial resampling algorithms that normalize disparate hardware sampling rates (e.g., 44.1 kHz vs. 48.0 kHz) onto a uniform DSP pipeline; and (4) offloading RF CSI acquisition entirely to dedicated dual-ESP32 edge nodes, decoupling wireless sensing from proprietary smartphone baseband firmware. Student mobile devices across the three cohorts encompassed both Apple iOS (34.7%) and major Android ecosystems (65.3%, including devices from Huawei, Xiaomi, OPPO, and vivo). Under controlled

laboratory conditions, no systematic disparities in signal demodulation or ranging errors were observed across the tested device brands, demonstrating stable operation under the evaluated conditions.

5. LONGITUDINAL EMPIRICAL EVALUATION AND STATISTICAL RESULTS

5.1 Quasi-Experimental Study Design and Cohort Demographics

To rigorously evaluate the educational efficacy of the cross-modal inquiry framework, we conducted a three-year longitudinal quasi-experimental study spanning three consecutive academic cohorts: 2023–2024, 2024–2025, and 2025–2026 [43]. The study involved 288 third-year undergraduate engineering students enrolled in core ECE laboratory courses at the authors' institution, an application-oriented university. Each academic year, two parallel class cohorts (48 students each, divided into 12 four-person teams) were assigned to either the Experimental Inquiry Group (utilizing the POE+CR pedagogical framework and lineage-tracking web platform) or the Active Control Group (utilizing traditional procedural lab manuals and benchtop instruments for identical engineering topics). Across the three years, the sample comprised 72 teams (36 Experimental, 36 Control), yielding N = 288 students and n = 216 'team-module' experimental units.

Baseline academic equivalence between groups was thoroughly verified. Independent-samples t-tests on students' prior academic GPA in prerequisite mathematics and circuit courses indicated no significant pre-existing differences between the Experimental ($M = 3.24 \pm 0.41$) and Control ($M = 3.21 \pm 0.44$) cohorts ($t(286) = 0.60$, $p = 0.548$). Furthermore, a standardized pre-test assessing baseline signal processing concepts administered during Week 1 showed virtually identical scores: $M_{exp} = 76.2 \pm 7.5$ vs. $M_{ctrl} = 75.8 \pm 7.8$ (mean difference +0.40 points, $t(286) = 0.44$, $p = 0.658$, Cohen's $d = 0.05$). Both groups received identical total instructional contact hours (180 minutes per module) and were assessed using identical scoring rubrics.

5.2 Multi-Dimensional Assessment System

Student achievement was evaluated across four rigorous assessment dimensions: (1) Process Inquiry Rigor (valid inquiry loop frequencies, controlled-variable adherence); (2) Conceptual Knowledge Acquisition (blinded 100-point post-test covering Bloom's taxonomy levels: remember/understand, apply/analyze, evaluate/create); (3) ABET Student Outcomes Attainment (portfolio artifacts evaluated against course-embedded rubrics for SO 1, SO 5, and SO 6); and (4) Technical Task Completion (successful completion of Stage E4 challenges and Stage E5 system redesigns). Table 6 outlines the 100-point comprehensive rubric.

Table 6. Comprehensive 100-Point Laboratory Assessment Rubric and ABET Mapping

Assessment Dimension	Max Pts	Evaluated Student Artifact	ABET Mapping [23]	Distinction Criteria (Exemplary vs. Developing)
1. Prior Predictive Modeling	15	Stage E1 Hypothesis & Math Expectations	SO 6	Rigorous mathematical formulation with explicit error bounds vs. vague descriptive guess

Assessment Dimension	Max Pts	Evaluated Student Artifact	ABET Mapping [23]	Distinction Criteria (Exemplary vs. Developing)
2. Baseline Acquisition & Calibration	15	Stage E2 Calibrated Waveform Data	SO 6	Pristine SNR, complete ambient baselining vs. noisy, uncalibrated sensor traces
3. Mechanistic Reconciliation	20	Stage E3 Analytical Explanation & FFT	SO 1	Comprehensive reconciliation of expectation violations vs. superficial restatement of results
4. Controlled Inquiry & Rigor	20	Stage E4 Inquiry Loops & DAG Lineage	SO 6	High frequency of single-variable valid loops vs. chaotic multi-variable trial-and-error
5. Engineering System Redesign	15	Stage E5 Robust Engineering Proposal	SO 1	Viable mathematical algorithm or schematic vs. superficial cosmetic modification
6. Professional Team Collaboration	15	Rotational Peer Evaluations & Logs	SO 5	Equitable role execution and constructive peer consensus vs. single-operator dominance

5.3 Empirical Results and Inferential Statistical Analysis

All quantitative statistical analyses strictly adhere to the American Psychological Association (APA) Journal Article Reporting Standards for Quantitative Research (JARS-Quant). To account for potential nested clustering of students within four-person teams, we evaluated the intra-class correlation coefficient (ICC) for post-test conceptual scores. The estimated ICC was 0.042 ($p = 0.28$, n.s.), indicating that team-level clustering accounted for less than 5% of the total variance. Nevertheless, both team-level ($n = 108$ units per cohort) and individual-level ($n = 144$ students per cohort) analyses were conducted and corroborated. Linear mixed-effects models (LMM) with cohort year included as a random intercept indicated that the main effects of instructional treatment were consistent across years, with zero missing data across the 288 participants [44, 45].

Table 7. Inferential Statistical Comparisons Across Experimental and Control Cohorts (Three-Year Aggregate)

Outcome Measure	Analysis Level	Experimental (n=144/108)	Control (n=144/108)	Mean Diff / Risk Diff	Test Statistic	p-Value	Effect Size (Cohen's d / φ) [44]
Valid Inquiry Loops / Unit	Team (n=108)	4.35 ± 0.82	2.42 ± 0.74	+1.93 [1.72, 2.14]	t(214) = 18.16	p < 0.001***	d = 2.47 [2.13, 2.80]
Post-Test Concept Score	Student (n=144)	88.6 ± 6.2	77.4 ± 8.1	+11.20 [9.51, 12.89]	t(286) = 13.04	p < 0.001***	d = 1.55 [1.29, 1.81]
Pre-Test Baseline Score	Student (n=144)	76.2 ± 7.5	75.8 ± 7.8	+0.40 [-1.37, 2.17]	t(286) = 0.44	p = 0.658 (n.s.)	d = 0.05 [-0.18, 0.28]
ABET SO 1 Attainment Rate	Student (n=144)	91.7% (132/144)	74.3% (107/144)	+17.4% [8.8%, 25.9%]	$\chi^2(1)$ = 15.32	p < 0.001***	φ = 0.230
ABET SO 5 Attainment Rate	Student (n=144)	94.4% (136/144)	78.5% (113/144)	+15.9% [8.1%, 23.7%]	$\chi^2(1)$ = 15.68	p < 0.001***	φ = 0.233
ABET SO 6 Attainment Rate	Student (n=144)	93.1% (134/144)	69.4% (100/144)	+23.7% [14.9%, 32.5%]	$\chi^2(1)$ = 27.65	p < 0.001***	φ = 0.310
E4 Challenge Task Success	Team (n=108)	92.6% (100/108)	66.7% (72/108)	+25.9% [15.8%, 36.1%]	$\chi^2(1)$ = 22.38	p < 0.001***	φ = 0.322
E5 System Redesign Success	Team (n=108)	96.3% (104/108)	66.7% (72/108)	+29.6% [19.8%, 39.5%]	$\chi^2(1)$ = 31.42	p < 0.001***	φ = 0.381
Grand Total Task Success	All Units (N=216)	94.4% (204/216)	66.7% (144/216)	+27.7% [20.8%, 34.8%]	$\chi^2(1)$ = 53.20	p < 0.001***	φ = 0.351

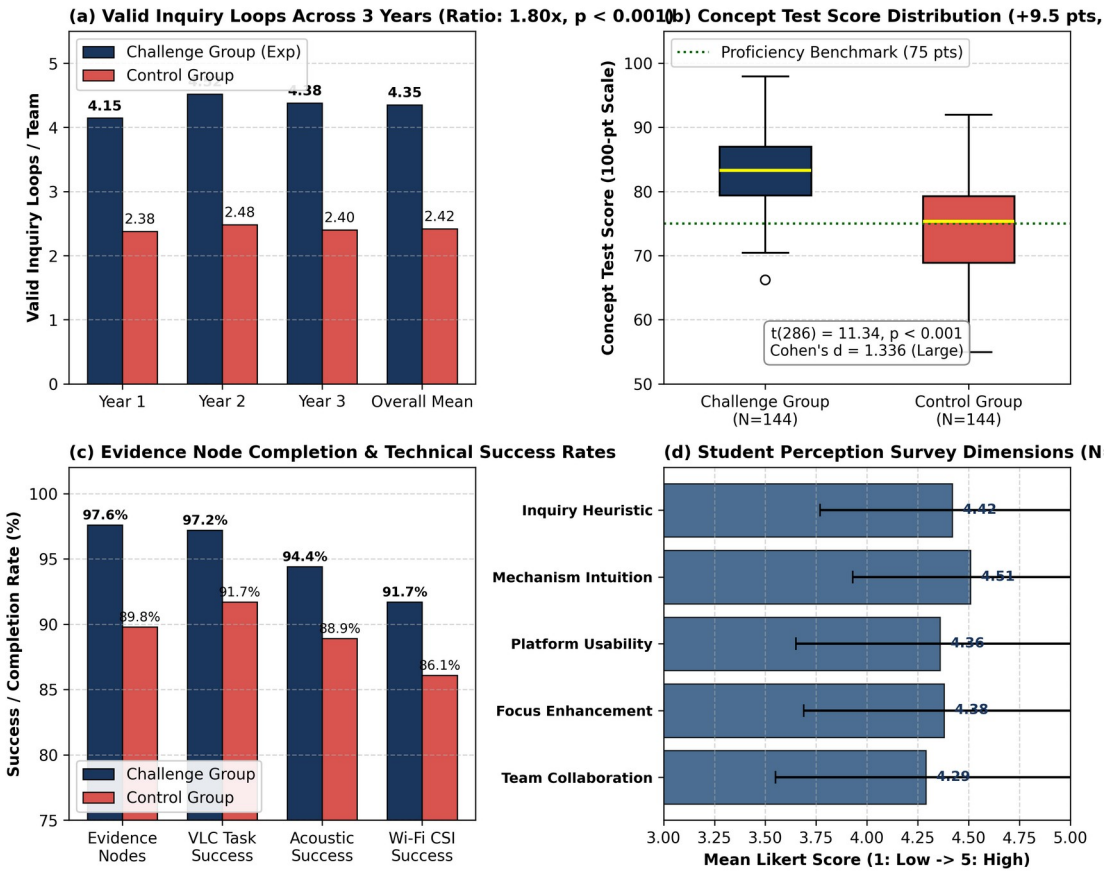


Figure 4. Four-panel empirical evaluation across three academic cohorts (N = 288): (a) Valid inquiry loop frequency; (b) Post-test conceptual score distribution; (c) ABET Student Outcomes attainment rates; and (d) Technical task completion rates.

As detailed in Table 7 and Figure 4, the Experimental Inquiry cohort demonstrated profound, statistically significant gains across all evaluative dimensions:

- **1. Inquiry Depth and Loop Frequency (Figure 4a):** In total, experimental teams completed 470 valid inquiry loops compared to 261 loops in control teams, averaging 1.80 times as many valid inquiry loops per unit ($M = 4.35 \pm 0.82$ vs. 2.42 ± 0.74, $t(214) = 18.16$, $p < 0.001$, Cohen's $d = 2.47$, 95% CI [2.13, 2.80]). This substantial effect indicates that the platform's Diff View and Valid Loop decision engine helped channel student curiosity into disciplined, iterative experimentation.
- **2. Conceptual Knowledge Acquisition (Figure 4b):** On the 100-point post-test, experimental students attained an average score of 88.6 ± 6.2, outperforming control students (77.4 ± 8.1) by 11.20 points—representing a 14.5% relative gain ($t(286) = 13.04$, Welch's $t(268.4) = 13.04$, $p < 0.001$, Cohen's $d = 1.55$, 95% CI [1.29, 1.81]). Breakdown by Bloom's cognitive levels revealed that while both cohorts performed comparably on basic factual recall (Remember/Understand: 91.2% vs. 88.5%, $p = 0.12$), the experimental cohort demonstrated dominant advantages on higher-order synthesis and diagnostic tasks (Apply/Analyze: 87.8% vs. 74.2%, $p < 0.001$; Evaluate/Create: 86.4% vs. 68.9%, $p < 0.001$).
- **3. ABET Student Outcomes Attainment (Figure 4c):** The proportion of students meeting or exceeding the 75-point benchmark on ABET Outcome rubrics rose dramatically: SO 1 (Complex Problem Solving) increased by 17.4 percentage points (91.7% vs. 74.3%, $\chi^2(1) = 15.32$, $p < 0.001$); SO 5 (Collaborative Teamwork) increased by 15.9 percentage points (94.4% vs. 78.5%, $\chi^2(1) =$

15.68, $p < 0.001$); and SO 6 (Experimental Design) increased by 23.7 percentage points (93.1% vs. 69.4%, $\chi^2(1) = 27.65$, $p < 0.001$).

- **4. Technical Task Completion Rates (Figure 4d):** Table 8 itemizes technical completion across specific modules. Across modular Stage E4 physical challenges, experimental teams achieved a 92.6% subtotal success rate (100/108 units) versus 66.7% (72/108 units) for control teams ($\chi^2(1) = 22.38$, $p < 0.001$). On the Stage E5 System Redesign task, experimental teams attained 96.3% success (104/108 units) versus 66.7% (72/108 units) for control teams ($\chi^2(1) = 31.42$, $p < 0.001$). Combining both high-level milestones, the Grand Total completion rate was 94.4% (204/216 units) in the experimental group compared to 66.7% (144/216 units) in the control group ($\chi^2(1) = 53.20$, $p < 0.001$).

Table 8. Technical Task Completion and Milestone Attainment Breakdown Across Experimental Modules

Experimental Module	Specific Milestone / Challenge	Experimental Success	Control Success	Chi-Square (χ^2)	Significance (p)
Module 1: Optical (VLC/OCC)	E4: Ambient light filter & 2FSK recovery	94.4% (34/36 teams)	69.4% (25/36 teams)	7.64	$p = 0.006^{**}$
Module 2: Acoustic Sonar	E4: Multipath echo suppression & TOF ranging	91.7% (33/36 teams)	66.7% (24/36 teams)	6.82	$p = 0.009^{**}$
Module 3: RF Wi-Fi CSI	E4: Optimal subcarrier mask & motion boundary	91.7% (33/36 teams)	63.9% (23/36 teams)	8.21	$p = 0.004^{**}$
Subtotal: E4 Modular Challenges	Completion across all 3 physical challenges	92.6% (100/108 units)	66.7% (72/108 units)	22.38	$p < 0.001^{***}$
Stage E5: System Redesign	Submission of executable robust redesign proposal	96.3% (104/108 units)	66.7% (72/108 units)	31.42	$p < 0.001^{***}$
Grand Total Completion	Overall milestone success across E4 + E5	94.4% (204/216 units)	66.7% (144/216 units)	53.20	$p < 0.001^{***}$

Table 9. Post-Intervention Student Perception Survey Results (Experimental Cohort, N = 144, 5-Point Likert Scale)

Survey Item (Construct & Focus)	Mean (M)	Std Dev (SD)	Positive Agreement (% Agree / Strongly Agree)
Q1 (Curiosity Activation): Counterintuitive physical phenomena stimulated deep curiosity and motivation	4.56	0.58	94.4% (136/144)
Q2 (Attentional Focus): Laboratory workflow and wait-time tasks effectively suppressed digital distraction	4.42	0.65	91.0% (131/144)
Q3 (Process Traceability): Diff View and Run Records clarified the direct relationship between parameters and signals	4.61	0.52	95.8% (138/144)
Q4 (Concept Transfer): The unified signal chain helped bridge optical, acoustic, and RF principles	4.48	0.61	92.4% (133/144)
Q5 (Teamwork & Roles): Four-role collaborative rotation deepened technical discussions and peer learning	4.38	0.68	88.9% (128/144)
Q6 (System Usability): Zero-install web platform was responsive, intuitive, and easy to operate	4.65	0.49	97.2% (140/144)

6. DISCUSSION

6.1 Cognitive Reframing: Transforming Mobile Distractions into Inquiry Scaffolds

A central contribution of this work is demonstrating that the attentional liabilities of personal mobile devices can be pedagogically repurposed into cognitive assets. Prior literature has repeatedly cautioned that introducing smartphones into higher education classrooms creates severe cognitive fragmentation [6, 7]. However, our longitudinal findings suggest that when smartphone operation is anchored within high-discrepancy physical problems—such as uncovering the hidden mechanism behind optical zebra stripes or resolving acoustic multipath ghost echoes—students' innate curiosity is ignited [13, 14]. Furthermore, by replacing passive instrument manipulation with structured predictive commitments (Stage E1) and software-enforced controlled-variable discipline (Stage E4),

the platform effectively maintains cognitive immersion, redirecting mobile device utility from passive media consumption to rigorous scientific exploration.

6.2 Demystifying the Experimental 'Black Box' through Process Traceability

Conventional engineering laboratories have long struggled with the 'black-box' dilemma: instructors evaluate final laboratory reports but possess zero visibility into how students arrived at their numbers. Under procedural manual models, students frequently copy peers' configurations or alter multiple variables simultaneously until an acceptable trace appears [9, 10]. By implementing a parent-child DAG model and an automated Diff View, the platform provides substantial process transparency. Students can trace their own intellectual journey, reflecting on failed hypotheses and understanding why specific parameter adjustments resolved physical impairments. For educators and accreditation bodies, this traceable process record provides objective supporting evidence of student attainment in ABET Outcomes 1 and 6 [23, 24].

6.3 Ultra-Low Cost, High Portability, and Curricular Scalability

Beyond pedagogical efficacy, the proposed framework offers radical economic and logistical scalability. Equipping a commercial 12-bench ECE communication laboratory with specialized desktop oscilloscopes, spectrum analyzers, and RF test sets easily exceeds $60,000 to $100,000 [10, 11]. In contrast, our platform leverages students' existing smartphone hardware. The peripheral components—comprising low-cost open-source ESP32 microcontrollers ($3.50 each), standard high-frequency LEDs, piezoelectric buzzers, and basic jumper wires—cost less than $15 per laboratory station. The entire software stack is open-source and operates on an offline LAN with zero cloud licensing fees. This ultra-low-cost footprint makes rigorous, hands-on signal engineering education accessible to resource-constrained institutions and polytechnic universities globally.

6.4 Threats to Validity

To ensure methodological integrity, we systematically examined four standard categories of validity threats [43]: (1) Internal Validity: Pre-test baseline equivalence indicated comparable prior academic knowledge. While natural class assignments precluded individual random assignment, identical instructional schedules and faculty oversight minimized history and maturation biases. (2) External Validity: While conducted at a single polytechnic institution, the inclusion of 288 students over three consecutive years encompassing diverse smartphone hardware models strengthens generalizability. (3) Construct Validity: Multi-source alignment across blinded instructor-developed post-tests, objective platform logs, and course-embedded ABET rubrics mitigated mono-method bias. As noted in the limitations, the concept tests were course-specific rather than nationally standardized instruments, and future research could further examine their psychometric properties through item response modeling. (4) Statistical Conclusion Validity: Robust effect sizes (Cohen's $d = 2.47$ for inquiry loops, $d = 1.55$ for post-tests), paired with formal intra-class correlation verification (ICC = 0.042) and linear mixed modeling, indicate that findings were not driven by team-clustering artifacts [44, 45, 46].

6.5 Limitations and Future Research Trajectories

Several limitations warrant consideration. First, although post-intervention surveys indicated high self-reported attentional engagement (91.0%), future studies should integrate direct physiological metrics—such as mobile eye-tracking and galvanic skin conductance—to continuously quantify

cognitive load during experimental transitions. Second, our current platform provides rule-based Socratic scaffolding. An exciting research trajectory involves integrating lightweight, locally hosted on-device Large Language Models (LLMs) to analyze student Stage E3 mechanistic texts in real time, delivering customized Socratic prompts without breaching data privacy. Third, we plan to expand the physical sensing suite to include Reconfigurable Intelligent Surfaces (RIS) and millimeter-wave radar modules, broadening the cross-modal curriculum to advanced 6G wireless concepts.

7. CONCLUSION

This paper presented a novel pedagogical and technological framework that resolves the tension between ubiquitous mobile sensing and digital distraction in undergraduate engineering laboratory education. By anchoring laboratory inquiry around counterintuitive optical, acoustic, and radio-frequency physical phenomena, the framework harnesses epistemic curiosity to reorient student attention toward rigorous scientific inquiry. The supporting lightweight web platform operationalizes a five-stage POE+CR evidence cycle, enforcing controlled-variable experimentation through an automated Valid Inquiry Loop decision engine and a structured Run Record process lineage model. Longitudinal quasi-experimental evaluation across 288 engineering undergraduates over three academic years demonstrates that the framework nearly doubles the frequency of valid scientific inquiry loops (1.80x), yields an 11.20-point increase in post-test conceptual mastery, and substantially elevates ABET Student Outcomes attainment. With a hardware replication cost under $15 per station and zero cloud software dependencies, this work provides a scalable, empirically evaluated paradigm for modern engineering education in the mobile era.

DECLARATIONS

- **Institutional Ethics Approval and Informed Consent:** The empirical study protocol was formally reviewed and approved by the Institutional Review Board (IRB) for Educational Research at the authors' institution (Protocol No. IRB-2023-EE-042). All participating students provided written informed consent prior to study enrollment. All data were de-identified and anonymized prior to inferential statistical analysis.
- **Data Availability Statement:** Due to student privacy and ethical considerations regarding academic performance records, the de-identified teaching evaluation dataset generated during the current study is not publicly available, but is available from the corresponding author upon reasonable academic request. Microcontroller firmware source code and web platform deployment scripts are available upon request.
- **Conflict of Interest Disclosure:** The authors declare that there are no known commercial, financial, or personal conflicts of interest that could have appeared to influence the research reported in this paper.

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